Bootstrap 5 Student Guide

A beginner-friendly, fully offline Bootstrap 5.3.3 website with a homepage, login, signup and dashboard page. Use it to study Bootstrap and as a starting template for your own project.

1. What is Bootstrap?

Bootstrap is a free collection of ready-made CSS and JavaScript. Instead of writing all the CSS yourself, you add class names to your HTML and Bootstrap styles it for you.

```
<button class="btn btn-primary">Click me</button>
```

"btn" and "btn-primary" are Bootstrap classes. They turn a plain button into a blue, rounded, styled button.

- Saves time: buttons, forms, menus and cards are already designed.
- Responsive: pages look good on phones, tablets and computers.
- Consistent: everything shares one look.

2. Quick Start

1. Download or copy this project to your computer.
2. Open the folder in VS Code (or any code editor).
3. Double-click index.html to open it in your browser. No internet or server is needed.
4. Click through the pages: Home, Login, Sign Up, Dashboard.
5. Change something in the code, save, and refresh the browser to see the result.

Tip: In VS Code, the Live Server extension refreshes the page for you automatically.

3. Folder Structure

File / Folder	What it is	Edit it?
index.html	Homepage	Yes
pages/login.html	Login page	Yes
pages/signup.html	Sign up page	Yes
pages/dashboard.html	Student dashboard	Yes
assets/css/style.css	Your custom styles	Yes
assets/css/bootstrap.min.css	Bootstrap styles	No
assets/css/bootstrap-icons.min.css and fonts/	Icon font	No
assets/js/bootstrap.bundle.min.js	Bootstrap JavaScript	No
assets/images/	Photos and favicon	Replace with your own
docs/	This guide	No

Rule of thumb: edit the HTML files and style.css. Do not edit files ending in .min.css or .min.js.

4. Key Words

Word	Simple meaning	Example
Class	A name you add to an HTML tag to style it	class="btn"
Component	A ready-made piece of UI (navbar, card, modal)	class="card"
Utility	A small class that does one job (spacing, color, alignment)	mt-3, text-center
Container	The box that holds page content and sets side margins	class="container"
Grid	A layout system of 12 columns per row	row and col-md-6
Breakpoint	A screen width where the layout changes	md = 768px and wider
Responsive	Layout adapts to the screen size	Columns stack on phones
CDN	A website that hosts Bootstrap files online. Not used here; we use local files.	-

The Grid in 30 seconds

```
<div class="container">
  <div class="row">
    <div class="col-md-6">Left half</div>
    <div class="col-md-6">Right half</div>
  </div>
</div>
```

- A row has 12 columns.
- col-md-6 means "take 6 of 12 columns (half) on medium screens and up".
- On small screens (phones), columns stack on top of each other.

Breakpoints

Prefix	Screen width	Device
(none)	under 576px	Phone
sm	576px and up	Large phone
md	768px and up	Tablet
lg	992px and up	Laptop
xl	1200px and up	Desktop

5. Pages and What to Study

Page	What it shows	Bootstrap features to study
index.html	Navbar, hero, features, courses, contact form, footer	navbar, container, row/col grid, cards, badges, forms, utilities (py-5, shadow-sm)
login.html	Split-screen login form	grid, input-group, form-check, buttons, responsive columns
signup.html	Registration form	form-select, nested grid, order-* classes, form-control
dashboard.html	Sidebar, stat cards, progress bars, table	offcanvas, dropdown, nav-pills, progress, table-responsive, badges

6. Step-by-Step: Build Your Own Project

Step 1: Copy the template

Copy the whole bootstrap-template folder and rename it (for example my-portfolio).

Step 2: Change the site name and title

- Change <title>EduBoost | Home</title> to your own title.
- Change EduBoost in the navbar and footer to your project name.

Step 3: Make a new page

6. Copy pages/login.html and rename it, for example pages/about.html.
7. Delete the content inside <body>, but keep the <script> line at the bottom.
8. Keep the <link> lines in <head>. They load Bootstrap.
9. Add your own content, or copy the navbar from index.html (fix its link paths).

Step 4: Check your file paths

Page location	CSS path
Main folder (index.html)	assets/css/bootstrap.min.css
Inside pages/	../assets/css/bootstrap.min.css

"../" means "go up one folder". Every page needs these lines (add ../ if the page is inside pages/):

```
<!-- In <head> -->
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="assets/css/bootstrap.min.css" rel="stylesheet">
<link href="assets/css/bootstrap-icons.min.css" rel="stylesheet">
<link href="assets/css/style.css" rel="stylesheet">

<!-- Right before </body> -->
<script src="assets/js/bootstrap.bundle.min.js"></script>
```

Step 5: Add a navbar link to your new page

```
<li class="nav-item"><a class="nav-link" href="pages/about.html">About</a></li>
```

Step 6: Add a component from the Bootstrap docs

10. Go to getbootstrap.com/docs/5.3/components/card/ (or any other component).
11. Copy the example code and paste it inside your <body>.
12. Save and refresh the browser.

Tip: Do not copy the CDN <link> or <script> lines from the docs. This project already loads Bootstrap locally.

Step 7: Change colors and style

- Use Bootstrap color classes such as btn-success, bg-warning, text-danger.
- For your own main color, edit the top of assets/css/style.css:

```
:root {
  --brand: #4f46e5;       /* your main color */
  --brand-dark: #3730a3;  /* darker shade for hover */
}
```

The btn-brand, text-brand and bg-brand classes use these colors.

Step 8: Replace images

13. Put your picture in assets/images/.
14. Update the src in your HTML:

```
<img src="assets/images/my-photo.jpg" class="img-fluid rounded" alt="Describe the picture">
```

img-fluid makes the image shrink to fit small screens. Always write an alt description.

Step 9: Test on different screen sizes

15. Open the page in Chrome or Edge and press F12.
16. Press Ctrl + Shift + M to turn on device mode.
17. Pick a phone or tablet size and check that nothing breaks.

7. Cheat Sheet: Where to Use What

Need	Use	Where
Center content with side margins	container	Wrap each page section
Full-width content	container-fluid	Dashboards, split screens
Columns	row + col-*	Inside a container
Space between columns	g-4	On the row

Spacing (scale 0 to 5)

- m = margin (outside space), p = padding (inside space).
- Sides: t top, b bottom, s left, e right, x left and right, y top and bottom.
- Examples: mt-3, p-4, py-5, mx-auto (centers a block).

Text, buttons and flexbox

Class	Result
fw-bold	Bold text
text-center	Centered text
text-muted	Gray text
display-4	Very large heading
lead	Larger intro paragraph
btn btn-primary	Solid blue button
btn btn-outline-secondary	Outline button
btn-lg / btn-sm	Bigger / smaller button
d-flex	Put items side by side
justify-content-between	Push items to opposite ends
align-items-center	Center items vertically
gap-2	Space between items
d-none d-md-block	Hide on phones, show on tablets and up

Icons

```
<i class="bi bi-house"></i>
```

Browse icon names at icons.getbootstrap.com.

9. Common Mistakes and Fixes

Problem	Likely cause	Fix
Page has no styling	Wrong CSS path	Check assets/ vs ../assets/
Icons show as empty squares	Icon CSS or fonts folder missing	Keep bootstrap-icons.min.css and css/fonts/ together
Navbar button or dropdown does nothing	JavaScript not loaded	Add bootstrap.bundle.min.js before </body>
Columns do not sit side by side	Missing row, or columns outside it	Use container > row > col-*
Layout is tiny on phone	Missing viewport tag	Add the meta viewport line
Change does not appear	File not saved or browser cache	Save, then press Ctrl + F5
Custom style is ignored	style.css loaded before Bootstrap	Load style.css after bootstrap.min.css
Images are missing	Wrong file name or path	Check spelling and capital letters

10. Helpful Links

- Bootstrap Documentation (start here): https://getbootstrap.com/docs/5.3/getting-started/introduction/
- Grid System: https://getbootstrap.com/docs/5.3/layout/grid/
- Components: https://getbootstrap.com/docs/5.3/components/
- Forms: https://getbootstrap.com/docs/5.3/forms/overview/
- Utilities: https://getbootstrap.com/docs/5.3/utilities/spacing/
- Bootstrap Icons: https://icons.getbootstrap.com/
- Bootstrap Examples: https://getbootstrap.com/docs/5.3/examples/
- Free photos: https://unsplash.com/

11. Credits

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